

NORTHSTAR RETAIL DISTRIBUTION

Submitted to Cobalt Ridge Automation in support of warranty claim WR-2026-0417

Claim evidence submission

Submitted 20 April 2026 by D. Whitfield, Director of Facilities · Joliet Distribution Center

Per your request of 17 April, attached is the documentation we hold covering the SR-440 sort line and the work performed on it. This is everything our maintenance team was able to locate. Please advise if anything further is needed.

This file contains several separate documents submitted together and was not split before sending.

Warranty claim intake · inbound message

Captured from the Cobalt Ridge support portal, case WR-2026-0417

FromTobias Okafor <t.okafor@northstarretail.example>
Tosupport@cobaltridge.example
CcDana Whitfield <d.whitfield@northstarretail.example>
SentFriday, 17 April 2026 06:58 CDT
SubjectURGENT - SR440 down at Joliet, we are losing the outbound wave
We need help this morning. The main sort line stopped at about 6:40 and we cannot get it to run. It faulted out during the peak outbound wave, so we have trailers staged and nothing moving to them. We are hand sorting what we can but that is maybe fifteen percent of rate and we will start missing carrier cutoffs by ten o'clock.

Operators say it threw the same overtemp fault we have been seeing on and off, then went into a hard stop and would not reset. Third reset attempt it faulted again inside two minutes.

This line is under warranty, it went in September 2024. We need somebody on site today. What I need to know right now is when a technician can be here and whether there is anything my team can safely do in the meantime to get partial flow back.

For context on our exposure, a full day down at this building is on the order of two hundred seventy five thousand dollars in expedite and penalty costs, and this is our heaviest week of the month.

Dana is copied and can approve whatever is needed on our side.

Tobias Okafor
Maintenance Supervisor, Joliet Distribution Center
Northstar Retail Distribution
Direct (815) 555-0193 · Mobile (815) 555-0271

From: Tobias Okafor
Sent: Tuesday, 7 April 2026 16:22 CDT
Subject: FYI - drive work on SR440

Heads up, we had our maintenance contractor swap the main sort drive yesterday. We had been getting overtemp faults for a couple weeks and your lead time was three weeks which we could not absorb going into peak. They put in an equivalent unit and got us running. Let me know if you need anything for your records.

Attachments (3): SR440_fault_screen_0641.jpg (1.2 MB) · staged_trailers_0705.jpg (2.8 MB) · NRD_asset_register_extract.pdf (340 KB)

Intake classification

Field	Recorded value
Case	WR-2026-0417
Received	17 April 2026, 06:58 CDT
Channel	Support portal, email intake
Asset	Sortation Line SR-440, Joliet DC
Impact tier	Tier 1, Line Down
Stated exposure	\$275,000 per day, customer's own figure, unverified

Warranty status at intake	Active, expires 14 September 2026
Prior related correspondence	7 April 2026, third-party drive replacement notified after the fact

Illustrative document created for the FUSION 2026 demonstration. All names, addresses, and domains are fictional.

MERIDIAN INDUSTRIAL SERVICES

Contract Maintenance · Material Handling · 4471 W Dixie Hwy, Bolingbrook IL 60440 · (630) 555-0148

FIELD SERVICE WORK ORDER

W/O NUMBER: MIS-44182

DATE OF SERVICE: 04/06/2026

CUSTOMER: Northstar Retail Distribution

PO REF: NRD-MNT-2026-0331

SITE: Joliet DC, Bldg 2

ARRIVED: 07:15 DEPARTED: 15:50

EQUIPMENT: Sortation line, "SR440"

TECHNICIAN: R. Vasquez, Lead Tech

CALLED IN BY: T. Okafor, Maint Supv

BILLING: T&M, contract rate

SERVICE CLASSIFICATION (check all that apply)

☒ Corrective repair

☐ Preventive maint

☒ Parts replaced

☐ Warranty work

☒ Billable T&M

☒ Controls modified

☐ OEM authorized

☐ Full validation done

☒ Customer approved on site

REPORTED CONDITION

Customer reports intermittent stoppages on main sort line, 3-4 events per shift over prior two weeks. Maint supv states drive is "running hot" and throwing overtemp faults during peak. Requests we look at it before peak season. Customer indicates OEM lead time was quoted at 3 weeks and they cannot wait.

WORK PERFORMED

Checked drive enclosure temps at three points during run. Recorded 71C, 74C, 78C at power stage housing under sustained load. Ambient in aisle was 29C. Drive fault log showed repeated thermal cutback events. Visual on the existing unit shows discoloration on the heatsink face.

Advised customer the drive should be replaced. OEM unit not available in required window per customer. Customer authorized substitution with equivalent unit from our stock. Installed unit as follows:

Position	Removed	Installed
Main sort drive	CR-DRV-4410-B (Rev B), s/n 4410B-22781	Altek AD-5500-HD, s/n AD55-91043

Substitute unit is rated at same HP and voltage. Frame and mounting pattern required a 6mm adapter plate, fabricated on site. Unit is not a drop-in but fits within the existing guard envelope.

CONTROLS ADJUSTMENTS

Replacement drive would not hold the existing accel profile without tripping on the first divert wave. Adjusted the following parameters to get the line running:

Parameter	Was	Set to	Reason
ACCEL_RAMP_MS	420	610	Reduce inrush on start
CURRENT_LIM_PCT	115	135	Prevent nuisance trip at divert

THERM_CUTBACK_C	68	82	Substitute unit runs warmer by spec
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Line ran clean for 90 min observed after adjustment. Customer maintenance witnessed startup.

TESTING / VALIDATION

Performed run-in at operating speed for 90 minutes. Did NOT perform full OEM commissioning test sequence. We do not have the OEM test fixture or the current baseline document on file for this line. Recommended customer obtain sign-off from the OEM. Customer maintenance supervisor indicated they would "handle that side of it."

TECHNICIAN NOTES

Told the supervisor twice that raising the thermal cutback is a stopgap and the root cause of the original overheating was never established. He acknowledged. Recommend OEM review before peak. Also the aisle has poor airflow, two of the four ceiling fans in that bay are not running.

PARTS & LABOR

Item	Qty	Amount
Altek AD-5500-HD drive assembly	1	\$11,480.00
Adapter plate, fabricated	1	\$340.00
Labor, lead technician, 8.5 hr	8.5	\$1,487.50
TOTAL		\$13,307.50

Technician · R. Vasquez

Customer acceptance · T. Okafor

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COBALT RIDGE AUTOMATION

Installed Base Records · Asset Service History Extract

Service history · Sortation Line SR-440

Asset ID

SR-440

Serial

CRA-SR440-2113-0087

Customer

Northstar Retail Distribution

Site

Joliet Distribution Center, Building 2, Outbound

Commissioned

14 September 2024

Warranty status

Active, expires 14 September 2026

Baseline

CR-SR440-3.2 (controls set 2025.11)

Service visits

6

Open findings

1

Extract generated

17 April 2026

Visit log

Visit 1 · 22 September 2024 · Commissioning handover

Cobalt Ridge · Technician: J. Whitfield

Full commissioning sequence completed and signed. Throughput verified at 11,400 units per hour sustained. Baseline CR-SR440-3.1 released to site. Customer maintenance team trained on daily checks.

Visit 2 · 3 March 2025 · Scheduled preventive maintenance

Cobalt Ridge · Technician: J. Whitfield

Schedule B tasks completed. Belt tension adjusted on infeed. No findings.

Visit 3 · 19 September 2025 · Scheduled preventive maintenance

Cobalt Ridge · Technician: A. Nakamura

Schedule B tasks completed. Two divert photo-eyes replaced under consumables. Controls set updated to 2025.11 and baseline reissued as CR-SR440-3.2. Customer signed acceptance of the revised baseline.

Visit 4 · 18 January 2026 · Scheduled preventive maintenance

Cobalt Ridge · Technician: A. Nakamura

Schedule B tasks completed. **Finding recorded and left open:** main sort drive case temperature measured 64C under sustained load, against a nominal expectation of 52C to 58C for this duty. No fault codes present and no line stop observed during the visit. Advised site to monitor and to check bay ventilation. Recommended a follow-up thermal survey within ninety days.

Finding reference: SF-2026-0114. Status at extract: OPEN. No follow-up visit recorded.

Visit 5 • 6 April 2026 • Third-party work, recorded from customer notification

Not a Cobalt Ridge visit • Recorded 9 April 2026 from customer email

Site advised that a contract maintenance provider replaced the main sort drive and adjusted controls parameters. No prior authorization request was received by Cobalt Ridge for this work. Copy of the third-party work order requested from the customer on 9 April. Not received as of this extract.

Note added by A. Nakamura: this work was not performed under the service agreement and the installed component has not been qualified against the baseline.

Visit 6 • 11 April 2026 • Remote evidence review

Cobalt Ridge • Reliability and Controls Engineer: M. Alvarez

Remote pull of controls configuration against released baseline CR-SR440-3.2. Three parameters differ from the released values. Differences correspond to the period around 6 April. No approval record exists for the changes. Flagged to Aftermarket Operations for warranty review.

Summary

Category	Count	Comment
Scheduled maintenance visits	3	All Schedule B tasks completed on interval
Corrective visits by Cobalt Ridge	0	No covered repair performed to date
Third-party interventions recorded	1	Drive replacement, 6 April 2026, unauthorized
Open findings	1	SF-2026-0114, elevated drive temperature, January 2026

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